



SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: CSA1407

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS:

Total Marks:50

Annexure A

SCENARIO BASED TASK

Code of Conduct:

I M S GOWRISH SHANKAR REDDY (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M S GOWRISH SHANKAR REDDY

Reg No: 192424270

1). Parsing Table Conflict in LL(1):

In an LL(1) parser, the parsing table is constructed using the FIRST and FOLLOW sets. The FIRST set identifies the terminals that can begin strings derived from a production, while the FOLLOW set identifies terminals that can appear immediately after a non-terminal.

A conflict occurs when more than one production is entered into the same parsing table.

Decision:

Modify the grammar by eliminating left recursion, performing left factoring, and ensuring the FIRST and FOLLOW set do not create conflicts. This produces a conflict-free LL(1) grammar.

2). Incorrect Parse Tree Generation:

For the expression $id + id - id$, the syntax analyzer constructs a parse tree by applying grammar productions to the input tokens. The structure of the tree determines

the order in which the operations are evaluated.

The problem occurs when the grammar doesn't properly define operator associativity.

$id + (id - id)$

Instead of the correct left-associative grouping:

$(id + id) - id$

Decision:

Use grammar rules that enforce left associativity for addition and subtraction:

$E \rightarrow E + T \mid E - T \mid T$

$T \rightarrow id$

3. Invalid Expression Handling:

The lexical analyzer correctly identifies a , $+$, and b as valid tokens.

However, the syntax analyzer checks whether these tokens occur in a valid grammatical arrangement.

The expression $a + b$ is invalid because after the $+$ operator, the grammar expects an operand, but the next token is another operator $*$. Therefore the token sequence violates the grammar and a valid parse tree can't be generated.

Decision:

Use error recovery techniques such as panic mode recovery, phrase-level recovery, and synchronization using follow sets. These techniques allow the parser to continue processing the remaining input while giving meaningful errors.

4. Ambiguity in Conditional Statements

The grammar is:

$$S \rightarrow \text{if } E \text{ then } S$$
$$S \rightarrow \text{if } E \text{ then } S \text{ else } S$$
$$S \rightarrow a$$

This grammar is ambiguous b/c the parser cannot determine which if statement an a belongs to.

Decision:

Rewrite the grammar by separating matched and unmatched statements.

$$S \rightarrow M \mid U$$

$$M \rightarrow \text{if } E \text{ then } M \text{ else } M_1$$

$$U \rightarrow \text{if } E \text{ then } S \mid \text{if } E \text{ then } M \text{ else } U.$$

Alternatively, a parser can follow the common rule that an else is associated with the nearest unmatched if.

5. Operator precedence conflict

The grammar is

$$E \rightarrow E + E$$

$$E \rightarrow E * E$$

$$E \rightarrow \text{id}$$

This grammar is ambiguous because it doesn't specify operator precedence or associativity. For the input:

$$\text{id} + \text{id} * \text{id}$$

two possible parse trees are:

$$(\text{id} + \text{id}) * \text{id} \quad \text{and} \quad \text{id} + (\text{id} * \text{id})$$

Decision:

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow \text{id}$$

RUBRICS FOR CALCULATION:

Scenario based assessment

Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand